INCH-POUND

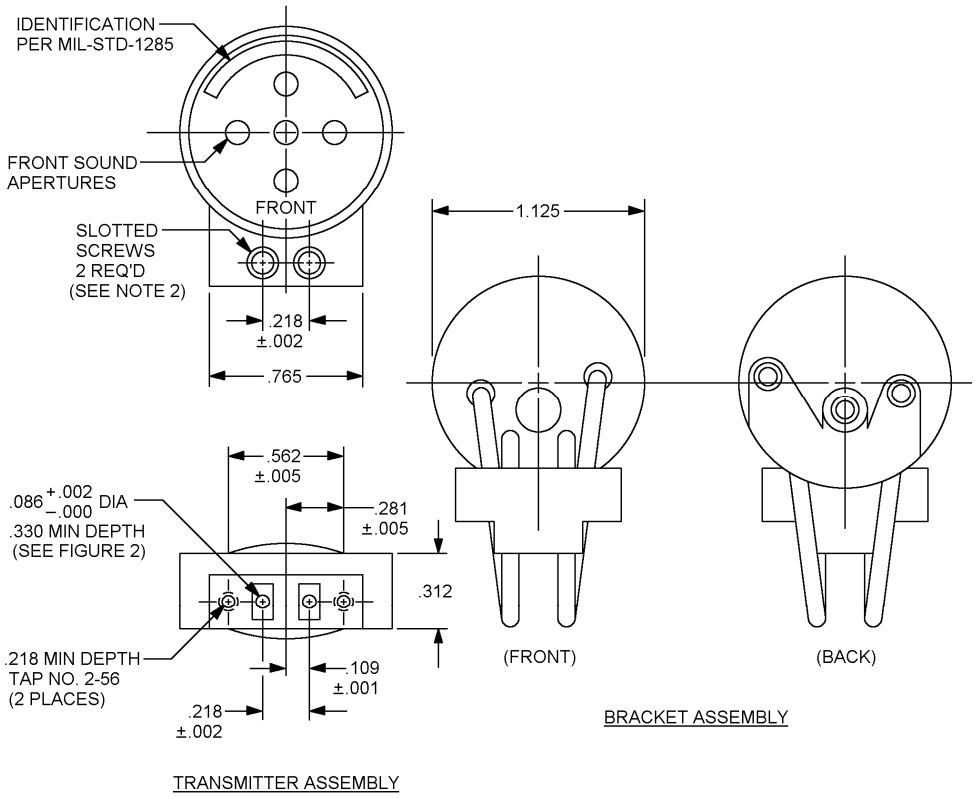
MIL-PRF-26542/7B
16 May 2005
SUPERSEDING
MIL-M-26542/7A
w/AMENDMENT 1
20 March 2002

PERFORMANCE SPECIFICATION SHEET
MICROPHONE, M-94B/A

Inactive for new design after March 12, 2002.

This specification sheet is approved for use by all
Departments and Agencies of the Department of Defense.

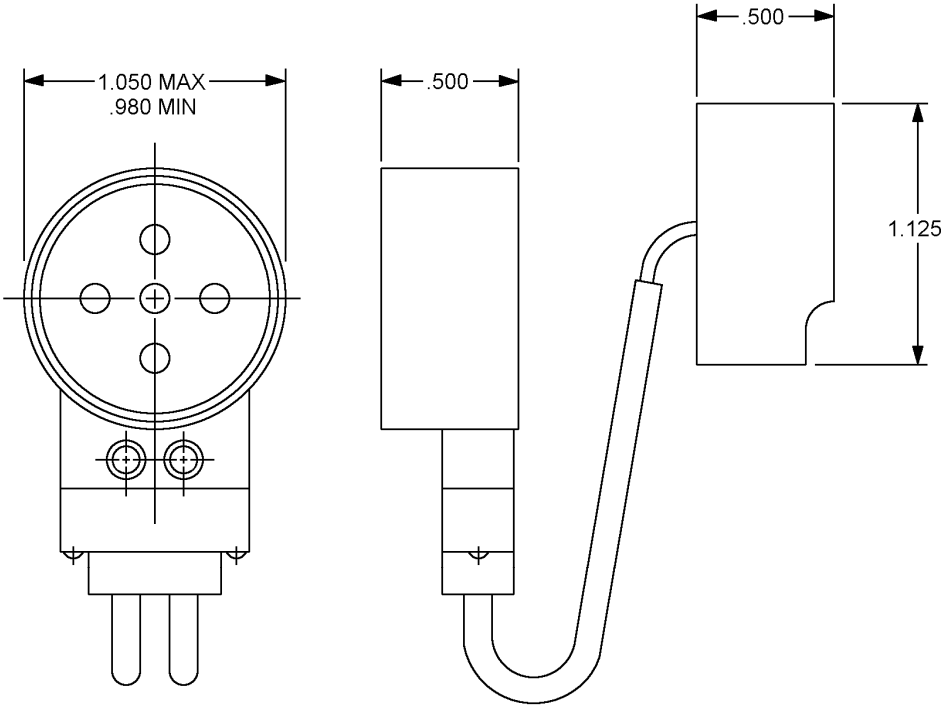
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-26542.



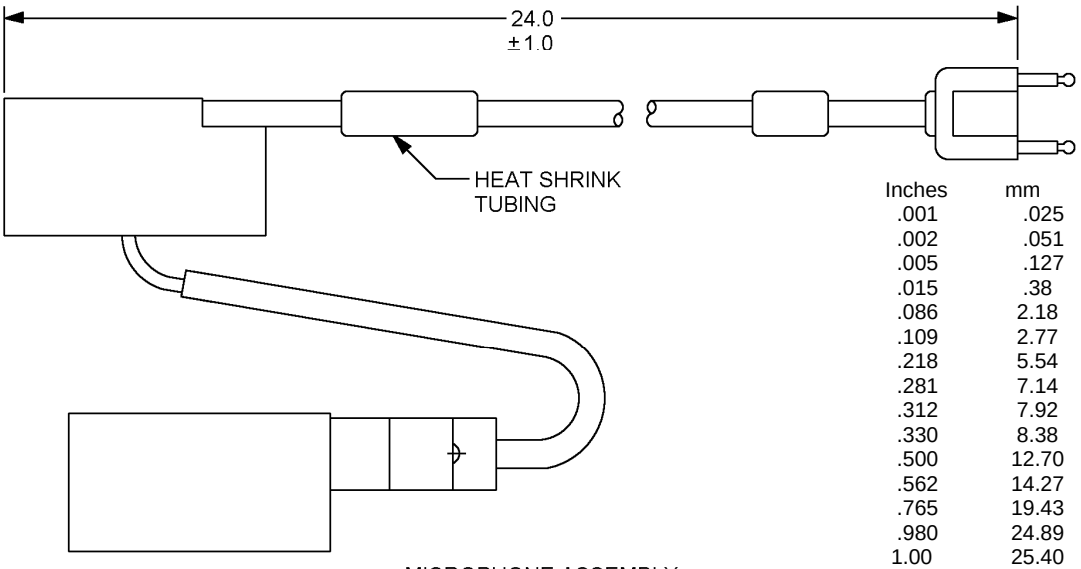
NOTES:
1. Dimensions are in inches. Unless otherwise specified, tolerances are ± .015 (0.38 mm).
2. Set-screws shall hold the part securely to the bracket, shall be slotted (for interchangeability of spares among manufacturers and tri-service applications), and shall be recessed.
3. Hole depths shall be as specified in order to provide a secure bracket connection, which does not compromise the mechanical stability of the microphone base.
4. Sound aperture location and configuration are optional, provided that frequency response, environmental and performance requirements are met as specified.
5. Metric equivalents are given for general information only.

FIGURE 1. Microphone assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.



TRANSMITTER AND BRACKET ASSEMBLY

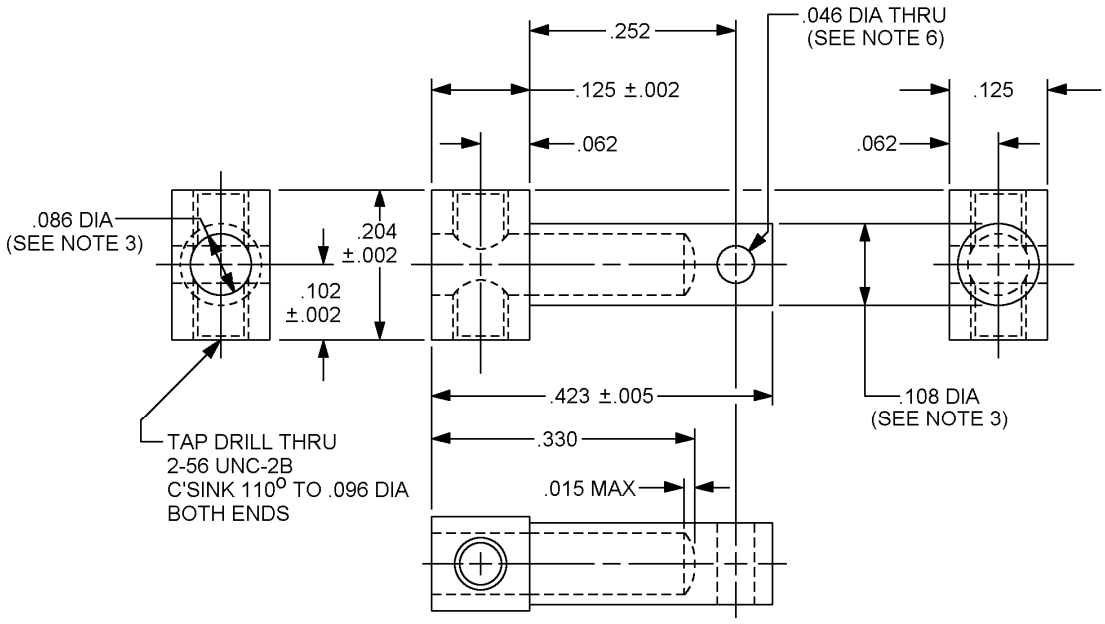


MICROPHONE ASSEMBLY
(SIDE VIEW)

Inches	mm
.001	.025
.002	.051
.005	.127
.015	.38
.086	2.18
.109	2.77
.218	5.54
.281	7.14
.312	7.92
.330	8.38
.500	12.70
.562	14.27
.765	19.43
.980	24.89
1.00	25.40
1.050	26.67
1.125	28.58
24.00	609.60

FIGURE 1. Microphone assembly (Continued).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.

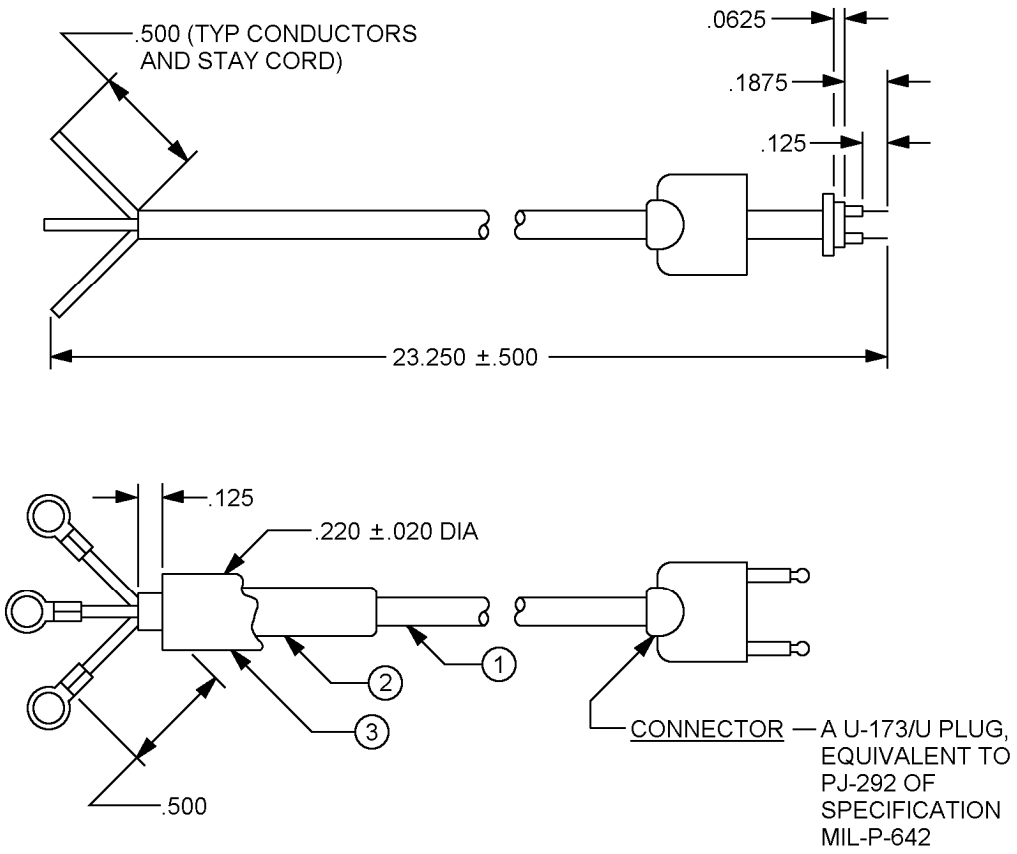


Inches	mm	Inches	mm
.002	0.05	.102	2.59
.003	0.08	.108	2.74
.005	0.13	.125	3.18
.015	0.38	.204	5.18
.046	1.17	.252	6.40
.062	1.57	.330	8.38
.086	2.18	.423	10.74

NOTES:
1. Material to be brass in accordance with ASTM B 16, temper H02 (Half Hard).
2. Silver plate finish in accordance with ASTM B 700, Grade A (MAT, without luster).
3. Diameter to be concentric within .005 inch, Total Indicator Reading (T.I.R.).
4. Dimensions are in inches. Unless otherwise specified, tolerances are ± .003 inch.
5. Metric equivalents are given for general information only.
6. Solder point connection design is at the option of manufacturer.

FIGURE 2. Connector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.

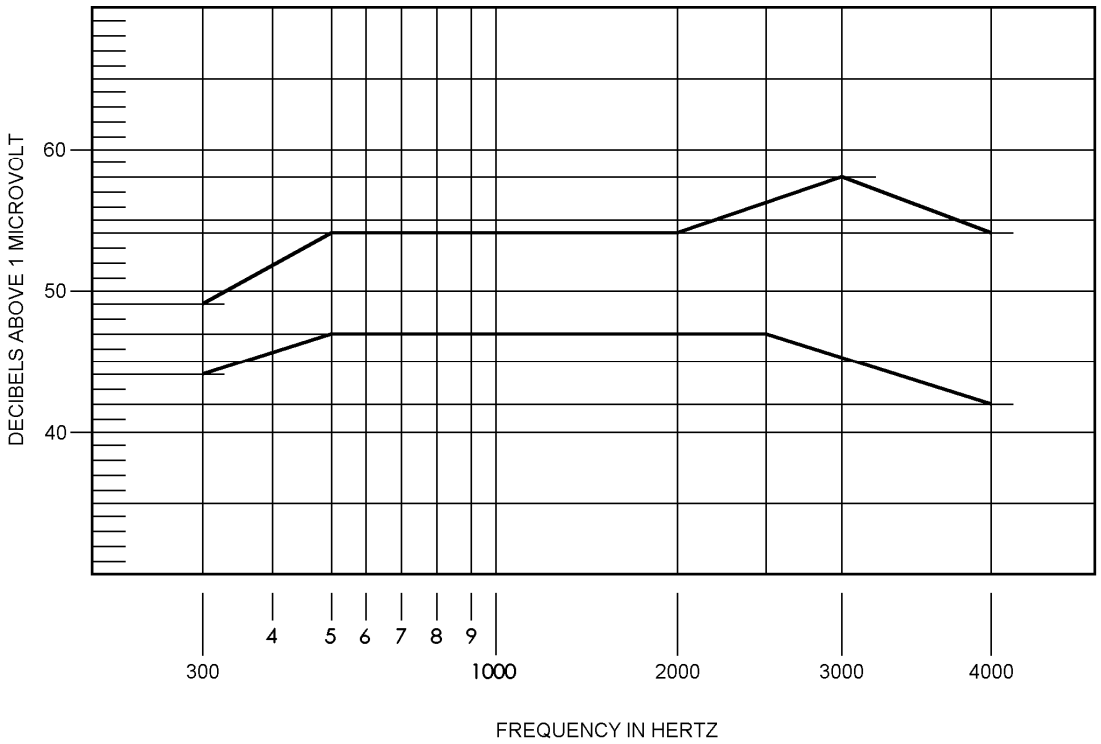


Inches	mm	Inches	mm
.015	0.38	.1875	4.76
.020	0.51	.220	5.59
.0625	1.59	.500	12.70
.125	3.18	23.250	590.55

NOTES:
1. Heat shrink item 2 on item 1, then heat shrink item 3 onto item 2. Caution: Do not heat shrink item 2 and 3 at the same time.
2. Attach stay cord terminal, as close as possible to end of outer jacket.
3. Dimensions are in inches. Unless otherwise specified, tolerances are ± .015 inch.
4. Metric equivalents are given for general information only.

FIGURE 3. Cord assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.



Frequency points	300 Hz	500 Hz	2,000 Hz	2,500 Hz	3,000 Hz	4,000 Hz
upper limits (dB)	49.0	54.0	54.0	56.2 1/	58.0	54.0
lower limits(dB)	44.0	47.0	47.0	47.0	45.1 1/	42.0

1/ dB limits between key break point are calculated.

FIGURE 4. Frequency response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Color: Semi gloss black, 27038 in accordance with FED-STD-595.

Component parts: See figure 1.

Microphone element: See figures 1 and 2.

Bracket assembly: See figure 1.

Cord assembly: See figures 1 and 3.

Impedance: 4-6 ohms.

Resistive load: 5 ohms.

Performance:

Sensitivity at ground level: 51.5 ± 3 dB above 1 microvolt (μV), with a Sound Pressure Level input of 115 dB above 20 micropascals (μPa) at 1 kHz, when tested with the microphone sound port .187 inch ± .015 inch (4.75 mm ± 0.38 mm) from, and coaxial with, the opening of the artificial voice.

Sensitivity at simulated altitude: Sensitivity shall be not more than 3 dB below the ground level sensitivity, when tested at 10,000 feet.

Frequency response: The frequency response of the microphone, compared to the response of 51.5 dB at 1,000 Hz, shall be within the limits shown on figure 4, when tested with the microphone sound port .187 inch ± .015 inch (4.75 mm ± .38 mm) from, and coaxial with, the opening of the artificial voice. However, the difference between any two successive readings 100 Hz apart (up to 1,000 Hz) and 250 Hz apart (above 1,000 Hz) shall not exceed 2.5 dB, under all standard and service conditions. The response shall fall off at a rate of not less than 3 dB per octave, below 300 dB, and 6 dB per octave above 4,000 Hz.

Distortion: The harmonic distortion in the microphone shall not exceed 4 percent, under standard conditions, or 5 percent, after being subjected to service conditions.

Pressure equalization: The response of the microphone at 1,000 Hz shall not vary by more than 4 percent, from the reference established at 15,000 feet, while the microphone is subjected to a pressure increase corresponding to a descent from an altitude of 15,000 feet (16.7 inches of mercury) to sea level (30.0 inches of mercury) at a rate of 5,000 feet per minute.

Stray magnetic field: The stray magnetic field of the microphone shall not cause more than a 3º deflection, when placed within 8 inches of a magnetic compass.

Weight: The weight of the microphone shall not exceed 1.5 ounces.

Interchangeability: The equipment should meet the interchangeability requirements, as defined in MIL-HDBK-5400.

First article and conformance inspection: The microphone assembly shall be tested in accordance with the tests listed in table I. For First Article Test (FAT), see table I. For conformance inspection, see groups A and B in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.

First article inspection: FAT shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units, which have been produced with equipment and procedures normally used in production. Unless extended by the Government to other contracts or purchase orders, first article approval is valid only on the contract or purchase order under which it is granted.

Sample size: Six (6) microphones shall be subjected to first article Inspection.

Inspection routine: The sample shall be subjected to the inspections specified in table I, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into two sets of 3 (three) units each. One set shall be subjected to the inspections of group II. The other set shall be subjected to the inspections of group III.

Failures: More than one failure in group I and any failure in group II or III shall be cause for refusal to grant first article approval.

TABLE I. Parameter applicability.

Inspection[1/]	First article	Group A	Group B
Group I			
Visual and mechanical inspection	X	X	
Sensitivity at ground level	X	X	
Sensitivity at altitude	X		
Frequency response at ground level	X	X	
Frequency response at altitude	N/A		
Impedance	X	X	
Noise cancellation characteristics	N/A		
Effect of external magnetic field	N/A		
Stray magnetic field	X		X
Linearity	X		X
Talk-out	N/A		
Dielectric withstanding voltage	N/A		
Signal-to-noise	X		
Distortion	X		
Interchangeability	X		X
Group II			
Thermal shock	X		
Humidity	X		
Drop	X		
Pressure equalization	X		
Explosive decompression	N/A		
Salt fog	X		
Group III			
Vibration	X		
Bounce	N/A		
Altitude	X		
Moisture barrier seal	X		
Immersion	N/A		

1/ See the associated requirement and method paragraphs within sections 3 and 4 of MIL-PRF-26542, for the above inspections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.

Reference documents. In addition to MIL-PRF-26542, this document references the following:

ASTM B 16
ASTM B 700
FED-STD-595
MIL-HDBK-5400
MIL-P-642
MIL-STD-1285

The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5965-0382-000)

Review activities:
Army - AR, AT, AV, CR4
Navy - AS, OS
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:54Z
Check the source to verify that this is the current version before use.